

Progress Report — Technical Checkpoint

Controllable Text Generation via Diffusion Models

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1 Project Title, Group Members, and Problem Definition

Title. *Controllable Text Generation via Diffusion Models* (Term Project Topic #4).

Group members. Zübeyr Almaho (300201023) — single-member group.

Task and motivation. The project addresses **data-to-text generation with attribute control**: given a structured *meaning representation* (MR) listing restaurant attributes (e.g., `name[Loch Fyne]`, `food[Japanese]`, `priceRange[high]`, `familyFriendly[no]`), the system must produce a fluent natural-language description that faithfully realizes every requested attribute. This is a *conditional natural-language generation* task. The problem is important because controllable generation is a recurring bottleneck for safe deployment of large language models in product descriptions, dialog responses, report generation, and accessibility tooling: practitioners need outputs that remain faithful to a structured intent rather than drifting into hallucination or omitting required content.

Input / output. Input: a string of slot–value pairs (the MR). Output: a single English sentence (or short paragraph) realizing all slots.

Task type. *Generation* (conditional, data-to-text).

Why technically challenging. Autoregressive (AR) decoders tend to (i) under-explore the output distribution when forced through likelihood-maximizing decoding such as beam search, hurting *lexical diversity*, and (ii) silently drop or paraphrase slots when no explicit constraint enforces them, hurting *controllability*. Diffusion language models operate in a continuous embedding space and admit plug-and-play attribute control through inpainting or classifier guidance, but training them stably for short, structured text is non-trivial: schedule choice, rounding, sampling step count, and conditioning all interact.

Topic / title change. None since the proposal stage.

2 Dataset / Benchmark Status

Benchmark. E2E NLG *cleaned* [11], the sanitized release of the original E2E challenge corpus [10].

Source. Direct CSV download from the official cleaned release at <https://github.com/tuetschek/e2e-cleaning> (initially we used the `e2e_nlg_cleaned` HuggingFace dataset, but `datasets>=3.0` removed script-based loaders; we now fetch the canonical CSVs directly).

Scale and splits.

Split	MR-reference pairs	Unique MRs
Train	33,525	4,862
Validation	4,299	547
Test	4,693	630

Input format. Eight slot types: `name`, `eatType`, `food`, `priceRange`, `customerRating`, `area`, `familyFriendly`, `near`. We flatten each MR into a "key: value | key: value" prompt string and additionally keep the parsed slot dictionary for controllability evaluation.

Output format. A single English target utterance per training pair.

Preprocessing completed. (i) CSV download & on-disk cache at `data/raw/e2e/`; (ii) regex-based MR parser producing the slot dictionary; (iii) prompt flattening; (iv) model-specific tokenization: GPT-2 BPE for the GPT-2 baseline, the T5 SentencePiece tokenizer for T5, and the bert-base-uncased WordPiece tokenizer for Diffusion-LM; (v) prefix/target packing for the diffusion model (fixed prefix length $P=32$ for the MR, target length $L - P=64$).

Data issues encountered. (a) The `datasets>=3.0` script-loader deprecation, mitigated as above; (b) mild MR/reference noise in the original release, which the *cleaned* version already addresses; (c) strong class imbalance on the `familyFriendly` slot (binary, $\sim 50/50$, fine) and `customerRating` (skewed toward high ratings), which we will note in the error analysis section of the final paper.

Status. Dataset is fully downloaded, cleaned, parsed, and ready for experiments. The data loader is exercised by an automated smoke test (Section 10).

3 Literature Review Progress

We reviewed three threads of work and identified the methods that ground our choices of baseline, primary method, and evaluation.

(1) Diffusion language models. *Diffusion-LM* [1] introduced continuous-space text diffusion with a learned token embedding lookup and a transformer denoiser trained with x0-prediction plus a rounding cross-entropy. Its "sqrt" noise schedule and plug-and-play classifier guidance are central to our design. Earlier image-domain work — DDPM [2], improved DDPM [3], and DDIM [4] — provides the underlying forward process, the cosine schedule we use as an ablation, and the deterministic accelerated sampler we adopt at inference.

(2) Autoregressive conditional generation. GPT-2 [6] and T5 [7] are the standard AR references for the E2E benchmark. Most published E2E systems are encoder–decoder models or fine-tuned causal LMs; we therefore adopt one of each (GPT-2-small and T5-small) as baselines.

(3) Controllable generation and evaluation. PPLM [8] and FUDGE [9] establish gradient-based attribute control over AR decoders, motivating our comparison axis of *controllability* (slot accuracy). For evaluation we follow standard E2E practice: BLEU [12] and ROUGE-L [13] for fluency, *distinct-n* [14] for lexical diversity, and slot-fill accuracy for controllability.

Direction implied by the review. The literature supports our design: a Diffusion-LM-style continuous-text diffusion model with prefix conditioning is a faithful, well-motivated implementation of the topic, and the chosen baselines and metrics are the canonical comparisons for the E2E benchmark.

4 Baseline Models or Prompting Strategies

The project compares two AR baselines against the proposed *prefix-conditioned Diffusion-LM*. Both baselines are fully implemented end-to-end (training, generation, and evaluation).

Baseline 1 — GPT-2 (124M). Causal LM fine-tuning. Input format: "`<MR> <|sep|> <reference><eos>`"; labels equal inputs (standard causal-LM objective). Decoding: nucleus sampling ($p = 0.95$, $\tau = 1.0$). *Why suitable:* GPT-2 is the canonical decoder-only AR baseline for E2E NLG; nucleus sampling is the standard diversity-oriented decoding choice, giving the strongest possible *distinct-n* comparison for diffusion to match.

Baseline 2 — T5-small (60M). Encoder–decoder seq2seq with source = MR text, target = reference. Decoding: beam search ($b=4$). *Why suitable:* T5 represents the fluency-oriented AR family that traditionally dominates E2E leaderboards and provides the strongest *BLEU* reference point.

Implementation status. GPT-2 and T5 training pipelines are implemented with HuggingFace `Trainer` / `Seq2SeqTrainer` respectively, including per-epoch checkpointing, validation evaluation, and predict-with-generate. The training configurations are versioned at `configs/gpt2_baseline.yaml` and `configs/t5_baseline.yaml`.

Issues encountered. *Compute.* Local Mac hardware is insufficient for the diffusion run; we moved training to Colab Pro on an A100, which fits the full pipeline in one session. *Library compatibility.* HuggingFace `transformers` recently renamed `evaluation_strategy` → `eval_strategy`; the codebase targets ≥ 4.40 to remain forward-compatible. *API.* No paid API is used; everything runs locally / on Colab.

5 Initial Experimental Results

We report two classes of preliminary results at this checkpoint: (i) a quantitative *dataset characterization*, computed from the downloaded and parsed E2E NLG cleaned corpus, that establishes the diversity *ceiling* and length statistics our models need to match, and (ii) a first model comparison from the completed Colab Pro runs, summarized in the model-comparison table below and aggregated through `src/compare_results.py`.

Dataset characterization (computed locally from the cleaned CSVs).

Split	Pairs	Avg. ref. length	Vocab (refs)	distinct-1	distinct-2
Train	33,525	20.2 words	3,674	0.0054	0.0524
Validation	4,299	22.4 words	1,392	0.0144	0.0957
Test	4,693	24.1 words	1,399	0.0124	0.0802

Slot frequency on the training split (8 slot types, total 33,525 MRs): **name** 33,495 (99.9%), **food** 26,469 (79.0%), **eatType** 23,141 (69.0%), **priceRange** 20,692 (61.7%), **rating** 20,432 (60.9%), **area** 18,922 (56.4%), **familyFriendly** 17,706 (52.8%), **near** 14,103 (42.1%).

Interpretation. Three observations follow directly from these statistics and shape our modelling choices. First, the average reference length is ~ 20 – 24 words, so our diffusion maximum-length budget of 96 tokens (32 prefix + 64 target) is comfortably sufficient and we should not see truncation as a confound. Second, the natural distinct-2 of human references is only ~ 0.05 – 0.10 (higher on the smaller splits as expected from sample-size scaling); this is the *upper bound* that any well-behaved model can target without inventing implausible bigrams, which means a generator scoring close to this number is already approaching human-level diversity rather than being “too repetitive”. Third, **near** (42%) and **familyFriendly** (53%) are the least-frequent slots; we therefore expect these to be the most-dropped attributes in model outputs and we will analyse them specifically in the final error analysis.

Model comparison (Colab Pro, A100).

Model / strategy	BLEU	ROUGE-L	dist-2	Slot Acc.	Status
GPT-2 (nucleus)	46.4	57.2	0.17	0.82	Complete
T5-small (beam-4)	54.5	72.5	0.08	0.90	Complete
Diffusion-LM (proposed)	52.1	68.7	0.13	0.88	Complete

The pipeline itself has been end-to-end verified by a dedicated 10-second CPU smoke test exercising the MR parser, all three noise schedules, Diffusion-LM forward / backward / sample with prefix clamping, and the full metric pipeline. The smoke test confirms that gradients flow through the combined MSE-plus-rounding-CE loss and that the prefix is preserved at every denoising step. The completed main runs show a clear trade-off: T5-small is the strongest system on BLEU, ROUGE-L, and slot accuracy, GPT-2 achieves the highest lexical diversity, and Diffusion-LM lands between them with a better diversity–control balance than either extreme. In particular, Diffusion-LM improves *distinct-2* over T5 (0.13 vs. 0.08) while also improving slot accuracy over GPT-2 (0.88 vs. 0.82), which supports the project’s central motivation even though T5 remains the strongest pure accuracy baseline at this checkpoint.

6 Planned Improvements and Technical Direction

Beyond the current functional baseline of the diffusion model, we have identified four concrete improvements with high expected impact and tractable implementation cost:

1. **Self-conditioning** [5]. Feed the previous step’s $\hat{x}_0$ prediction as auxiliary input to the next step. Expected: $\sim+1$ –2 BLEU at no extra inference cost.
2. **EMA of denoiser weights.** Standard practice in diffusion training; expected: $\sim+0.5$ –1 BLEU and visibly smoother samples.
3. **Clamping trick at inference.** Periodically snap $\hat{x}_0$ to the nearest token embedding during sampling, as in the original Diffusion-LM paper; expected to improve both fluency and slot accuracy.
4. **Batched diffusion sampling.** Replace the current `batch_size=1` sampler with a batched version ($\sim 10\times$ speedup), which makes the full ablation suite feasible inside a single Colab session.

A stretch goal (treated as a graduate-level extension, listed as *future work* in the final paper if time does not allow) is to swap hard prefix clamping for *soft* classifier guidance: train per-slot classifiers on noisy embeddings and inject their gradients during sampling.

7 Ablation or Prompt Sensitivity Plan

The final paper will include a five-way ablation, with configurations already written and committed:

Axis	Variants	Hypothesis
Noise schedule	sqrt (default), cosine	cosine should hurt rounding accuracy at short lengths.
DDIM steps	50, 100, 200	quality saturates around 100 steps; cost–quality curve.
Prefix length	16, 32 (default), 48	longer prefix $\Rightarrow$ stronger control, less linguistic freedom.

DDIM-step ablations require *no* retraining (they reuse the base checkpoint); schedule and prefix-length ablations need separate training runs. The runner script `scripts/run_ablations.sh` executes the suite end-to-end and pipes results through `src/compare_results.py` to produce a single Markdown comparison table for the paper.

8 Error Analysis and Generation Quality Analysis

A dedicated Jupyter notebook (`notebooks/error_analysis.ipynb`) is already in place and will populate the final report. It computes: (i) per-slot *miss rate* (which attributes models tend to drop or paraphrase); (ii) output-length histograms per model; (iii) a diversity-vs-fluency scatter (distinct-2 against BLEU); (iv) the five worst-case per-example dumps for every model, ranked by per-example slot recall.

For the generation-quality discussion we will examine: *fluency* (does the model produce grammatical English?), *coherence* (does the sentence read as a unit?), *slot fidelity* (are all MR values present?), *repetition* (is there token-level or phrasal repetition? T5 with beam search is particularly susceptible), and *hallucination* (does the model invent attributes not in the MR? Free-form AR sampling is the typical offender). Representative cases for each failure mode will be extracted from the test set for the qualitative section of the LNCS paper.

9 Ethical Considerations and Bias

The project is a generative model and therefore inherits the standard set of responsible-AI concerns, in moderated form because the application domain (restaurant descriptions) carries low real-world harm potential.

Hallucination. AR baselines can invent attributes not in the MR; prefix clamping bounds this risk for the diffusion model but does not eliminate it in the free-form target region.

Dataset bias. E2E NLG is English-only, restaurant-specific, and United-Kingdom-centric; conclusions do not transfer cleanly to other languages or domains.

Evaluation limits. BLEU, ROUGE, and distinct- n are surface metrics and correlate imperfectly with human judgement of fluency. Our slot-accuracy metric is currently a strict substring match and will under-credit paraphrased attribute realizations; we plan to optionally integrate the official E2E scorer in the final paper as a sanity check.

Release. No model checkpoints will be released. The codebase is public to support reproducibility but the trained weights stay private to the project, sidestepping any misuse pathway.

10 GitHub and Reproducibility Status

Repository. <https://github.com/zubeyralmaho/diffusion-lm-ctg> (public).

Structure.

```
configs/           configs/ablations/      # YAML per experiment
src/data/          src/models/             # data loader, diffusion-lm, baselines
src/{train,generate,evaluate,compare_results}.py
scripts/{prepare_data,run_eval,run_ablations}.sh
notebooks/{colab_train,error_analysis}.ipynb
tests/smoke_test.py          # 10s CPU sanity test
report/{main.tex,progress_report.tex,refs.bib}
```

Implemented artifacts. data loader (CSV-based, self-caching); three training routines (GPT-2, T5, Diffusion-LM); unified generation script with model dispatch; evaluation script (BLEU, ROUGE-L, distinct-1/2, slot accuracy); five ablation configs; results aggregator; end-to-end Colab notebook with smoke-test gate; `HANDOFF.md` for collaborator-run training sessions.

Dataset preparation. `bash scripts/prepare_data.sh` downloads and caches the three E2E CSVs. No further preprocessing is required before training.

Reproducibility. `requirements.txt` pins the major versions (`torch>=2.1`, `transformers>=4.40`, `datasets>=2.18`, `accelerate>=0.30`, `sacrebleu>=2.4`, `rouge-score`, etc.). Every experiment

is reproduced from a single YAML config; no API keys or external services are used.

Commit history. The repository has consistent, descriptive commits spanning project scaffold, model implementation, prefix-conditioning support, ablation suite, error-analysis notebook, smoke test, dataset-loader fix, and this report. Each commit is small and topical, supporting the “progress monitored via GitHub commits” criterion.

11 Current Challenges and Next Steps

Current challenges.

1. *Compute capacity.* The Diffusion-LM training is the long pole (~ 45 min on A100, $\sim 2\text{--}3$ h on T4). Mitigation: Colab Pro session via a project collaborator, per-epoch checkpointing, smoke test before kickoff.
2. *Library churn.* `datasets>=3.0` removed script-loader support, which broke initial dataset access; resolved by switching to direct CSV download from the canonical source.
3. *Sampling throughput.* Diffusion sampling is currently non-batched, which is slow for the full test set; batched sampling is on the planned-improvements list.

Plan for remaining weeks.

- **Week 12 (current).** Submit this report. Main-model training and Section 5 result aggregation are complete; prepare the ablation jobs and final-paper tables.
- **Week 13.** Execute the ablation suite, complete the diversity-vs-fluency analysis, and add the four planned improvements listed in Section 6 (self-conditioning, EMA, clamping trick, batched sampling).
- **Week 14.** Error analysis (per-slot failure modes, qualitative case study, repetition / hallucination examples). Polish the LNCS paper draft and integrate all final tables and plots.
- **Week 15.** Final paper submission and the live demo (the same Colab notebook will serve as the demonstrable artifact).

Confidence assessment. The pipeline is implemented end-to-end, exercised by an automated test, and backed by completed main-run results; the remaining work is now ablations, deeper error analysis, and paper polishing rather than core model implementation, so we are on track for the final submission.

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References

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